MACHINE LEARNING PRACTICALS

Important Viva Questions and Answers | Short Revision Set

1. UBER FARE PREDICTION – LINEAR REGRESSION AND RANDOM FOREST

Q1. What is the aim of this practical?

Answer: To predict the Uber ride fare using ride details and compare Linear Regression with Random Forest Regression.

Q2. Is this classification or regression?

Answer: Regression, because the output fare_amount is a continuous numerical value.

Q3. What is the target variable?

Answer: fare_amount.

Q4. What are the input features used in the simple code?

Answer: Pickup longitude, pickup latitude, drop-off longitude, drop-off latitude and passenger_count.

Q5. What is preprocessing?

Answer: Preparing data before modelling, such as removing missing values and duplicates and checking invalid values.

Q6. What is an outlier and how is IQR used?

Answer: An outlier is an unusually high or low value. IQR is Q3 minus Q1; values below Q1 – 1.5×IQR or above Q3 + 1.5×IQR are flagged as potential outliers.

Q7. What is correlation?

Answer: Correlation measures the strength and direction of a linear relationship between numerical variables. It ranges from –1 to +1.

Q8. What is Linear Regression?

Answer: A supervised algorithm that predicts a numerical value by fitting a linear relationship between input features and the target.

Q9. What is Random Forest Regression?

Answer: An ensemble model that combines predictions from multiple decision trees, usually by averaging them.

Q10. What is R^2 score?

Answer: It measures how well the model explains variation in the target. A higher value is generally better; it can be negative.

Q11. What is RMSE?

Answer: Root Mean Squared Error measures the size of prediction errors in the same units as the fare. Lower is better.

Q12. How do you decide which regression model is better?

Answer: Compare both on the same test set. Generally, prefer higher R^2 and lower RMSE.

2. EMAIL SPAM DETECTION – KNN AND SVM

Q1. What is the aim of this practical?

Answer: To classify emails as spam or not spam using K-Nearest Neighbors and Support Vector Machine.

Q2. Is this classification or regression?

Answer: Binary classification, because there are two classes: spam and not spam.

Q3. What do the target values 0 and 1 mean?

Answer: 0 means not spam (normal email); 1 means spam.

Q4. What is the target column in the emails.csv file used here?

Answer: Prediction. This dataset uses Prediction, not label.

Q5. What are the input features in this dataset?

Answer: The numerical word-frequency columns. Email No. is an identifier and should not be used as a feature.

Q6. What is KNN?

Answer: K-Nearest Neighbors predicts a class using the majority class among the K closest training examples.

Q7. What does K mean in KNN?

Answer: K is the number of nearest neighbors considered when making a prediction.

Q8. What is SVM?

Answer: Support Vector Machine finds a decision boundary, called a hyperplane, to separate classes.

Q9. What are support vectors?

Answer: The training points closest to the SVM decision boundary; they help define that boundary.

Q10. Why scale features for KNN and SVM?

Answer: Scaling puts features on comparable ranges. It is especially important for KNN because it uses distance calculations.

Q11. What is a confusion matrix?

Answer: A table comparing actual and predicted classes. It shows correct and incorrect classifications.

Q12. What are accuracy, precision and recall?

Answer: Accuracy is the fraction of all predictions that are correct. Precision is the fraction of predicted spam that is truly spam. Recall is the fraction of actual spam correctly detected.

Q13. What is a false positive in spam detection?

Answer: A normal email incorrectly classified as spam.

Q14. Is TF-IDF needed for the provided emails.csv?

Answer: No. The version used here already contains numerical word-frequency features. TF-IDF is used when converting raw text into numerical features.

3. BANK CUSTOMER CHURN – ARTIFICIAL NEURAL NETWORK

Q1. What is the aim of this practical?

Answer: To predict whether a bank customer will leave the bank or stay using an Artificial Neural Network classifier.

Q2. Is this classification or regression?

Answer: Binary classification.

Q3. What is the target column and what do its values mean?

Answer: The target is Exited. 0 means the customer stayed; 1 means the customer left.

Q4. Which dataset should be used?

Answer: Churn_Modelling.csv, the bank customer churn dataset. Do not use sales_data_sample.csv.

Q5. Which columns should generally be removed?

Answer: RowNumber, CustomerId and Surname, if present, because they are identifiers rather than useful customer behaviour features.

Q6. Why convert Geography and Gender into numbers?

Answer: They are categorical text values. One-hot encoding, such as pandas get_dummies(), converts them into numerical columns.

Q7. What is an Artificial Neural Network (ANN)?

Answer: A machine-learning model made of connected artificial neurons that learn patterns from data.

Q8. What are the layers of an ANN?

Answer: Input layer receives features, hidden layer(s) learn patterns, and output layer produces the prediction.

Q9. What is MLPClassifier?

Answer: A scikit-learn implementation of a feed-forward Artificial Neural Network called a Multi-Layer Perceptron. It does not require TensorFlow.

Q10. What does hidden_layer_sizes=(6, 6) mean?

Answer: The network has two hidden layers, each containing six neurons.

Q11. What is an activation function?

Answer: A function applied to a neuron's input to introduce non-linearity so the network can learn complex patterns.

Q12. Why scale or standardize the data?

Answer: To put numerical features on comparable scales and help neural-network training.

Q13. What is a confusion matrix?

Answer: A table comparing actual and predicted classes. For two classes it shows true negatives, false positives, false negatives and true positives.

Q14. What was the result obtained in your practical?

Answer: Accuracy was 86.1%, with confusion matrix [[1529, 78], [200, 193]].

Q15. How many test customers were predicted to stay and leave?

Answer: 1,729 were predicted to stay and 271 were predicted to leave, out of 2,000 test customers.

Q16. Does 86.1% accuracy mean 86.1% of customers who left were detected?

Answer: No. Accuracy is overall correctness. The model correctly detected 193 of 393 actual churners, giving churn recall of about 49.1%.

4. DIABETES PREDICTION – KNN

Q1. What is the aim of this practical?

Answer: To predict whether a person belongs to the diabetes or non-diabetes class using KNN.

Q2. What is the target column?

Answer: Outcome. Usually 0 means no diabetes and 1 means diabetes; check the dataset's label definition.

Q3. What is KNN?

Answer: K-Nearest Neighbors predicts a class based on the majority class among the K closest training records.

Q4. What is Euclidean distance?

Answer: A straight-line distance measure between two points in feature space.

Q5. Why is feature scaling important for KNN?

Answer: KNN uses distances, so features with large numerical ranges could otherwise dominate the distance calculation.

Q6. What happens if K is too small?

Answer: The model may be sensitive to noise and overfit.

Q7. What happens if K is too large?

Answer: The model may become too generalized and miss local patterns.

Q8. What is a confusion matrix?

Answer: A table comparing actual and predicted classes, showing true negatives, false positives, false negatives and true positives.

Q9. What are accuracy, error rate, precision and recall?

Answer: Accuracy is correct predictions divided by total predictions. Error rate is 1 – accuracy. Precision is TP/(TP+FP). Recall is TP/(TP+FN).

Q10. What is a false negative in diabetes prediction?

Answer: The actual class is diabetes, but the model predicts no diabetes.

Q11. Why can accuracy alone be misleading?

Answer: If one class is much more common, a model can have high accuracy while missing many examples of the less common class. Precision and recall help show this.

5. SALES DATA CLUSTERING – K-MEANS AND HIERARCHICAL

Q1. What is the aim of this practical?

Answer: To group similar sales records using clustering and estimate the number of clusters with the Elbow Method.

Q2. Is clustering supervised or unsupervised?

Answer: Unsupervised learning, because there is no target label provided.

Q3. Which dataset is used and what features can be selected?

Answer: sales_data_sample.csv. Numerical features can include QUANTITYORDERED, PRICEEACH and SALES.

Q4. What is K-Means clustering?

Answer: An algorithm that divides data into K groups by assigning points to the nearest centroid and repeatedly updating the centroids.

Q5. What is a centroid?

Answer: The mean position of the data points assigned to a cluster.

Q6. What is the Elbow Method?

Answer: It runs K-Means for different K values and plots WCSS against K to help choose a suitable cluster count.

Q7. What is WCSS or inertia?

Answer: The sum of squared distances between each data point and its assigned cluster centroid. Lower values indicate tighter clusters.

Q8. Why does WCSS decrease when K increases?

Answer: More clusters allow points to be closer to their assigned centroids, so within-cluster variation generally decreases.

Q9. What is Hierarchical Clustering?

Answer: A clustering method that builds a hierarchy of nested groups.

Q10. What is Agglomerative Clustering?

Answer: A bottom-up hierarchical method that starts with individual data points and repeatedly merges the closest clusters.

Q11. What is the difference between K-Means and Hierarchical Clustering?

Answer: K-Means uses centroids and requires K to be specified. Hierarchical clustering builds a hierarchy by merging or splitting groups.

Q12. Why scale features before clustering?

Answer: Distance-based clustering can be dominated by features with larger numerical ranges. Scaling puts features on comparable ranges.

Q13. Is there a target column in this practical?

Answer: No. Clustering is unsupervised, so the algorithm creates cluster labels rather than predicting a known target.

Q14. Do cluster labels 0, 1 and 2 have a ranking?

Answer: No. They are arbitrary group identifiers. Examine each cluster's feature values to understand what it represents.

6. BASIC QUESTIONS ABOUT THE PYTHON CODE

Q1. What is pandas used for?

Answer: Reading, cleaning and manipulating tabular data.

Q2. What is NumPy used for?

Answer: Numerical calculations and working with arrays.

Q3. What does read_csv() do?

Answer: Loads a CSV file into a pandas DataFrame.

Q4. What does head() do?

Answer: Displays the first five rows by default.

Q5. What does shape return?

Answer: The number of rows and columns in the dataset.

Q6. What does isnull().sum() do?

Answer: Counts missing values in each column.

Q7. What does dropna() do?

Answer: Removes rows containing missing values by default.

Q8. What does drop_duplicates() do?

Answer: Removes duplicate rows.

Q9. What does train_test_split() do?

Answer: Divides the data into training and testing sets.

Q10. What does fit() do?

Answer: Trains the model using the supplied data.

Q11. What does predict() do?

Answer: Uses a trained model to generate predictions.

Q12. What does random_state=42 do?

Answer: Makes random splitting or initialization reproducible.

Q13. What does get_dummies() do?

Answer: Converts categorical columns into numerical indicator columns.

Q14. What does StandardScaler do?

Answer: Standardizes features to have approximately mean 0 and standard deviation 1.